



Who Connects Wikipedia? A Deep Analysis of Node Roles and Connection Patterns in Wikilink Network

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Abstract. Wikipedia is not only an online encyclopedia, but also a spontaneously formed network of knowledge. Its important information is contained not only in the content of the pages, but also in the links between them. The larger the network, the more complex its internal hierarchy. This study analyses the role of nodes in the Wikipedia link network and the linking patterns of the network, and finds that the network is composed of a few highly core nodes and a large number of edge nodes populating it. These high nodes are not tightly connected to each other, often in different knowledge communities, and are connected through a plurality of club nodes across communities, which play an integral role in the transfer and flow of information throughout the network. This study reveals the roles and information transfer mechanisms of key nodes in the network and provides a reference for building more efficient knowledge networks.

Keywords: Wikipedia · Complex Network · Community Structure · Diverse Club · Connection Patterns

1 Introduction

Knowledge innovation and growth are vital to human social and economic development, and the development of the information age poses new challenges to the integration of knowledge and the need for new forms of knowledge organization. As one of the world's largest self-organizing knowledge communities, Wikipedia's content entries are compiled collaboratively by volunteers and organized on the basis of group intelligence and a self-organizing development process. Wikipedia's reliance on group intelligence provides a model for the study of self-organized creation in the growth of scientific knowledge, and its linking structure and knowledge construction model are important for the construction and integration of group knowledge.

Lizorkin et al. divided the Wikipedia network into communities and found that there are semantically closely related entries between different communities in the network, which play an important role in connecting and communicating between different communities [1]. Brandes et al. analyzed the cooperation between editors by relationships, they found that the side relationships of the link network were related to the quality of

the lexical content [2]. Simulation of network dynamics [3] and growth models using the edge relations of Wikilink networks [4] is also one of the hot research topics, and these studies fully illustrate the vast amount of information hidden in the structure of Wikipedia link networks.

The information in Wikipedia's knowledge network is contained in the content of the entries and the links between them. The structure of the link network directly affects the efficiency of knowledge integration and the development pattern of Wikipedia. However, there is a lack of in-depth research on the structural form and formation process of Wikipedia's link networks. In order to grasp the mechanism of efficient organization of knowledge in the current information age, this study examines the changes in the key structure of Wikipedia link networks from the perspective of complex networks, using the English Wikipedia from 2002–2018 as the research object.

2 Methods

2.1 Nodes Connectivity in Community

The larger the network, the more complex the internal hierarchy. Guimerà and Amaral proposed a method for characterising the role of nodes in the context of the presence of a network community in 2005 [5]. The role of a node in a network is mainly determined by the strength of its connectivity within a community and the breadth of its connectivity between communities [5, 6], which are denoted using Z_i and P_i , respectively.

The within-module degree Z_i measures the strength of connectivity between node i and other nodes within its community.

$$Z_i = \frac{K_i - \bar{K}_{S_i}}{\sigma_{K_{S_i}}} \quad (1)$$

where K_{S_i} is the number of neighbouring nodes of node i in community s , $\bar{K}_{S_i}$ is the mean of the K values of all nodes in community s , and $\sigma_{K_{S_i}}$ is the standard deviation of the K values of all nodes in community s . From the above equation, Z_i is the quantile of z-score in the whole community and is the normalised level of the strength of the node's connection to the community within its community. Values generally range from -2 to 10, with nodes of higher degree generally having higher Z-scores.

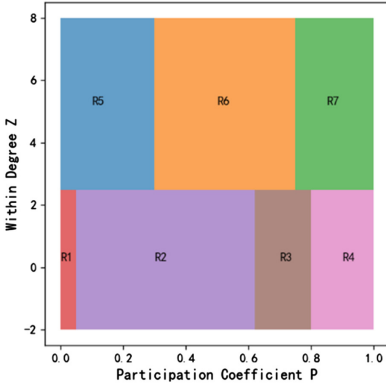
The participation coefficient measures the communication role of a node with other communities and ranges from 0 to 1. The more concentrated the edges of a node are in one community the smaller the value, and conversely the more evenly distributed it is across multiple communities the larger the value [5]. P can be expressed as

$$P_i = 1 - \sum_{s=1}^{N_M} \left(\frac{K_{is}}{K_i} \right)^2 \quad (2)$$

where N_M represents the total number of communities in the network, K_i is the degree of node i , and K_{is} the number of edges of node i in community s .

2.2 Network Role Segmentation

The P and Z values determine a node's connectivity role in its community and in other communities. Guimerà et al. proposed a method for classifying nodes' roles based on their P and Z values, and in this study nodes with Z values greater than 2.5 were classified as core nodes and nodes with Z values less than 2.5 were classified as non-core nodes with reference to the former. The specific division and its distribution in the Z - P parameter space are shown in the figure below. We can define the nodes in the wiki link network as different roles according to the above metrics, analyse their distribution and the connection methods between nodes in the Z - P space, and compare them with the random network to obtain the connection information between the nodes with different roles in the network [5] (Fig. 1)



(a) Regional distribution of node roles in Z - P space

$$\begin{aligned}
 Z \geq 2.5, & \begin{cases} P \leq 0.3, \text{ provincial hubs} \\ 0.3 < P \leq 0.75, \text{ connector hubs} \\ P > 0.75, \text{ kinless hubs} \end{cases} \\
 Z < 2.5, & \begin{cases} P \leq 0.05, \text{ ultra - peripheral nodes} \\ 0.05 < P \leq 0.62, \text{ peripheral nodes} \\ 0.62 < P \leq 0.8, \text{ non - hub connector} \\ P > 0.8, \text{ non - hub kinless nodes} \end{cases}
 \end{aligned}$$

(b) Node role division based on Z - P value

Fig. 1. Division of node roles in the network

2.3 Clubness of Complex Network

In order to describe the characteristics of nodes that are more connected to each other, the concept of rich-club was introduced by Colizza et al. in 2006 [7]. There exists a high degree of nodes in the network with strong connectivity between them, which can be measured by the rich-club coefficient, which is calculated as follows

$$\phi(k) = \frac{2E_k}{N_k(N_k - 1)} \quad (3)$$

$$\phi_{norm} = \frac{\phi}{\phi_{rand}} \quad (4)$$

where N_k is the number of nodes with degree greater than k and E_k is the number of connected edges between these nodes. The rich club factor is calculated by sorting the nodes in reverse order of their degree values, then removing them one at a time and

calculating the ratio of the number of contiguous edges in the network to the maximum possible number of contiguous edges. The rich club is a relative concept and to obtain a quantitative description requires generating a random network with the same degree distribution and dividing the two to obtain the normalised rich club coefficient.

In 2017 Bertolero et al. introduced the concept of the diverse-club [8], which measures the connectivity between nodes using a ranking of participation coefficients. A network with the phenomenon of diverse-club contains a group of nodes spanning multiple communities, connected to each other by multiple edges and simultaneously having a high level of mediated centrality. This is calculated as follows:

$$\theta(p) = \frac{2E_p}{N_p(N_p - 1)} \quad (5)$$

$$\theta_{norm} = \frac{\theta}{\theta_{rand}} \quad (6)$$

where N_p is the number of nodes with participation coefficients greater than P and E_p is the number of connected edges between these nodes. Similar to the rich club calculation process, the multivariate club coefficient is a function of the participation coefficient P . The nodes are sorted from smallest to largest P values, and the ratio of the number of remaining connected edges to the maximum number of possible connected edges in the network is removed and calculated, and the values are compared with a random network of the same size.

3 Data Pre-processing and Network Construction

Cristian et al. crawled and processed Wikipedia data from complete dumps prior to that year in 2019 to obtain highly formatted Wikipedia link network data by parsing and regular matching of xml files to obtain node and link data [9]. This dataset was used in this study to obtain link data from the establishment of Wikipedia in 2001 until March 1st, 2018. The dataset contains a complete snapshot of the Wikipedia link network for each year.

Based on the network created after data cleaning and removal of redirected links, this study analysed the relationship between the number of nodes and the number of edges of the English Wikipedia link network. We find that the nodes and edges of the linked network show an approximate linear growth over time. According to the study, the relationship between the number of edges $E(t)$ and the number of nodes $N(t)$ in a time-evolving network can be expressed as $E(t) \propto N(t)^\alpha$ [10][11], where $1 \leq \alpha \leq 2$. When $\alpha = 1$ the network shows a sparse growth pattern, at which time the new connected edges of the network cannot maintain the link strength between the new nodes and the average degree of the network remains unchanged; when $\alpha = 2$ the network is a fully connected network, where the added nodes are connected to all previous nodes; $1 < \alpha < 2$ corresponds to the case in between.

Plotting the number of nodes versus the number of edges for each year of Wikipedia to obtain Fig. 2, it can be found that Wikipedia shows a highly fitted linear growth relationship with $\alpha = 1$, at which point the network will become increasingly sparse

over time. The slope of the line with the redirected node data removed is 11, i.e. every additional 1 node in the network is connected to 11 nodes in the network, while in the original data this number is 1.8. It can be seen that about 80% of the connected edges in Wikipedia are redirected connected edges.

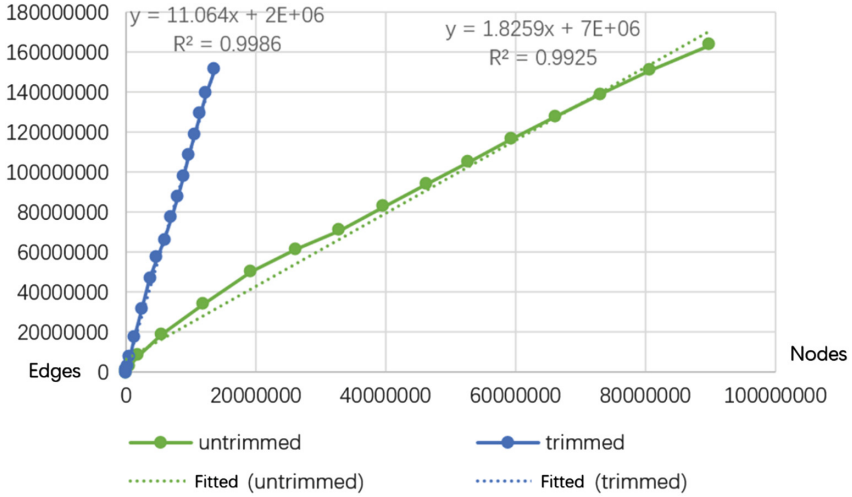


Fig. 2. Growth between the number of edges and nodes in Wikilink network

4 Node Role Classification and Distribution Characteristics

The underlying information is not sufficient for a complete portrayal of the network properties. In this section we take the 2002 network as an example, and will analyse the role and distribution of nodes in the Wikipedia link network. In this paper, the nodes in the network are divided into seven categories, and a scatter plot and density plot of the distribution of nodes in the linked network in 2002 are plotted in Z-P space, as shown in figure (a) below. The blue line in the figure represents the density distribution of nodes, the red on the right and upper side represents the density distribution of node attributes on each axis, and the grey bars and curves are the distribution box plots and density curves. We also labelled the nodes for each role in the network, as shown in figure (b) below. Nodes with a degree less than 20 have been removed from this figure, and the percentage marked is the proportion of nodes with a degree greater than 20 in the network (Fig. 3).

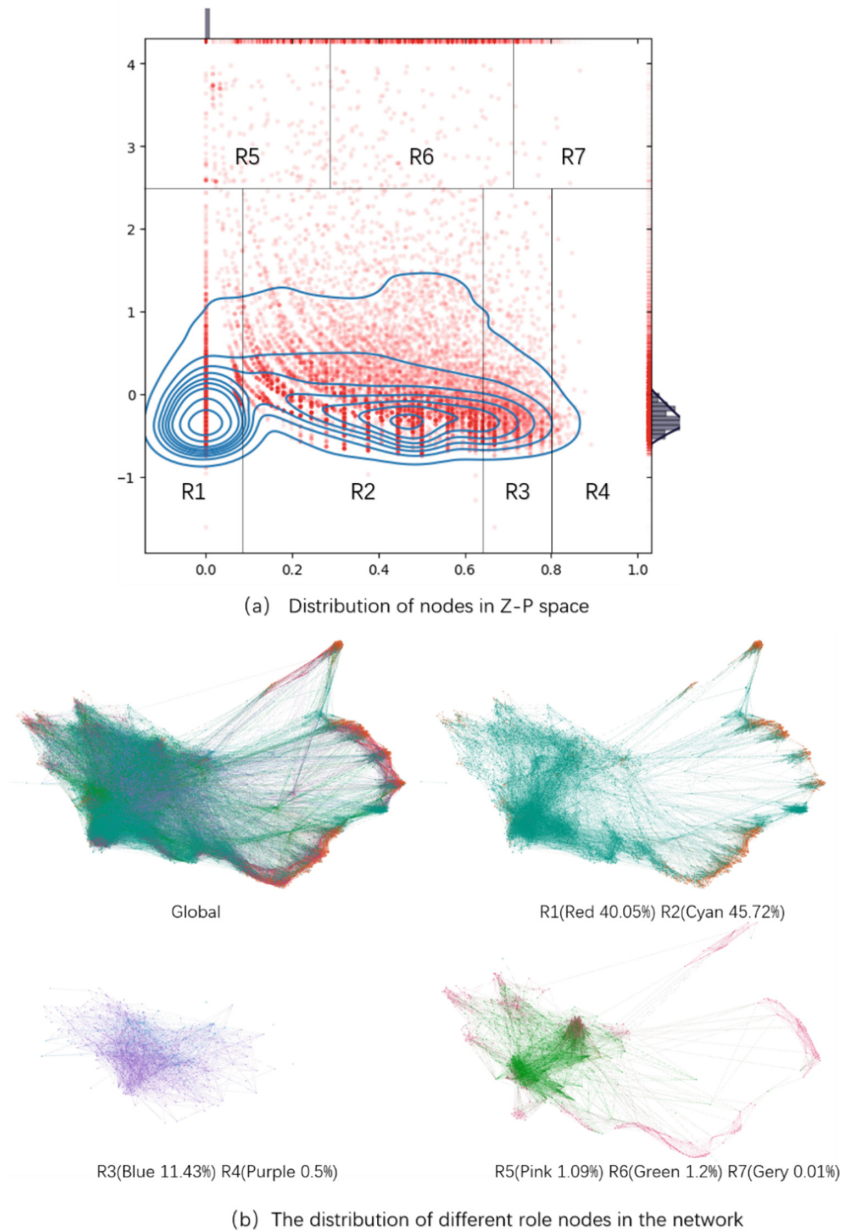


Fig. 3. Node role classification and distribution in 2002 Wikilink network

According to the density map we can find that the number of high Z-value R5-R7 nodes in the network is relatively small, while the low Z-value R1 and R2 types are relatively small. In terms of distribution, the edge nodes in the network show a roughly concentrated distribution and have less impact on the overall shape of the link

network, but the percentage of these nodes in the network determines the overall size of the network. Combining the proportions of the different roles and their distribution in the network, we believe that the R3, R4, R5 and R6 nodes in the network mainly influence the overall connectivity characteristics of the network. The core nodes rely on their height values to connect a large number of nodes in their own communities, and have extensive links to other communities in the amateur world. The distribution of the height-valued nodes can be seen to be highly overlapping with R1 and R2 nodes and are centrally located. These core nodes are connected to other communities in the network, and are not particularly prominent in their own participation coefficients due to their height values, but their location determines the important role they play in the connectivity of the network. The high p-values R3 and R4 nodes are a series of bridge nodes with low degree values but widely present across communities, as can be seen from the distribution, and they are widely present across the clusters R1 and R2 nodes. In other words, the whole linking network is a network skeleton composed of core nodes with high Z-values. Their high values ensure the linking strength of the core structure of the network, and the rest of the nodes with high P-values populate the network skeleton to keep the different knowledge communities closely connected to each other, constituting a complex and ponderous knowledge network.

5 Clubness of Wikilink Network

In order to further analyse how nodes with high Z-value and high P-value are connected in the network, this paper analyses the club phenomenon in the network, and calculates the clubness coefficient in the network after removing nodes according to the percentage size of degree, P-value and Z-value respectively, and the results are shown in the figure below. It can be found that there is no obvious rich club and Z-club phenomenon in the network, and even the remaining link strength in the network is lower than that of a random network of the same size after the percentage of deleted nodes exceeds 80%. This may be related to the fact that the high-value and high-Z-value nodes are scattered in different knowledge communities in the network; for example, it is difficult for core nodes in the humanities to form connections with high-value nodes in the astronomy community, and although there may be more links between the two communities, these connections do not exist between the smaller number of high-value and high-Z-value nodes. This may be due to the fact that height-valued nodes in the network belong to separate domains of biology, for example anatomy and plants are difficult to directly relate to each other, so the connections between height nodes are not strong. A corresponding explanation can also be obtained from the division of node attributes in the Z-P space, where a higher proportion of core nodes are local core nodes, and such nodes tend to be connected only to nodes within the community, with fewer connections between local core nodes in different communities, again demonstrating that the connections between height nodes in the network are not strong. Due to the high correlation between node Z-values and degrees, rich clubs and Z-clubs show similar results (Fig. 4).

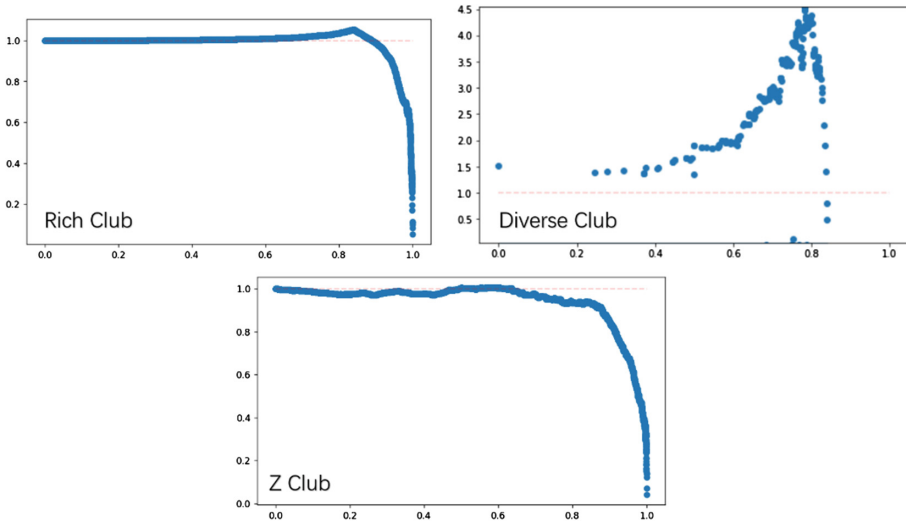


Fig. 4. The clubness coefficients for various clubs in 2002 Wikilink network

On the other hand, when nodes are removed in order of participation coefficient, the strength of connections in the network is consistently significantly higher than in a random network of the same size, with the remaining network being about four times stronger at 90% removal, after which an inflection point occurs, due to the large number of nodes with zero participation coefficients, whose neighbours are located in the same community and do not have connected edges across communities. After removing these nodes the remaining part of the network is very tightly linked. The existence of the plural club phenomenon suggests that nodes spanning different communities play an important role in linking knowledge in Wikipedia, mainly referring to R3, R4 and R7 nodes in the node classification. Combining the distribution of nodes with high participation coefficients in the network, as indicated in Fig. 3.11, with the distribution of nodes in the R3 and R4 categories in Fig. 3.12, we find that these nodes are in a position between communities and play an important role in the connectivity of the communities.

6 How Nodes of Different Role are Connected

Following an analysis of the connectivity patterns between the global nodes of the network, this study analyses the connectivity patterns between the various types of nodes in the network. This study proposes a random network generation algorithm that maintains the same distribution of network neutrality and modularity for random networks for comparison.

The number of nodes of each role in the network was first counted and compared with the random network generated by the same degree distribution and the algorithm in this paper, where each type of random network was generated 10 times to reduce the error and the average was taken as the final result, as shown in the figure below. When comparing the two, it can be seen that the number of nodes in networks with the same

degree distribution can vary depending on the structure and nature of the network. The number of R3 and R4 nodes in the random network is significantly higher than that of the original and reconstructed networks, while the number of R1 and R2 nodes is lower. This is due to the fact that random networks are selected with equal probability from among all nodes of the network when performing edge rearrangement, whereas the algorithm used in this paper is a random selection of nodes within and across communities in which they are located, placing restrictions on the connectivity range connections of knowledge lexemes that are more realistic (Fig. 5).

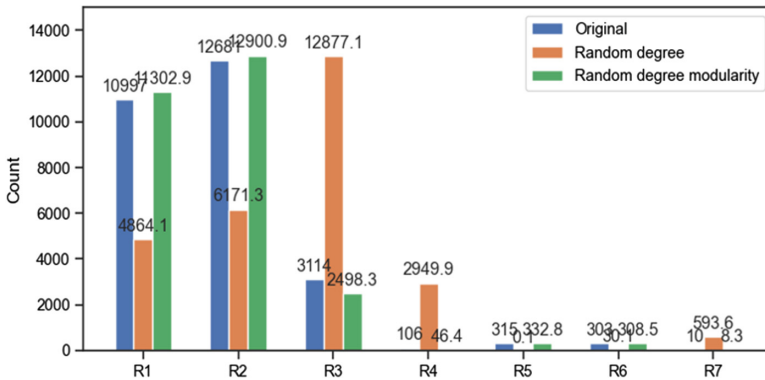


Fig. 5. Number of nodes for different roles in 2002 Wikilink network

The paper then analyses the average degree of the neighbouring nodes of each role node, as shown below. In order, the original network in 2002, the random network in igraph and the network generated by the algorithm used in this paper. The horizontal axis is the degree value of each type of node, the vertical axis is the average degree of neighbouring nodes, and each line represents the different role nodes plotted in double logarithmic coordinates. It can be noticed that the results of the original network are relatively similar to those of the networks generated in this paper, while the results of the random network show another pattern. Specifically the results for the network as a whole are broadly similar due to the same degree distribution for all three networks, but the results for the nodes of each role are quite different. In the random network R1, R2 and R3 nodes all preferred to connect to nodes with higher degree values, showing some heterogeneity. Whereas in the wiki network the tendency for R1 role nodes to connect to high degree nodes is closer to the network overall, high participation coefficient nodes show a similar tendency, and overall there is little variability in the tendency to link across role nodes with different degree values (Fig. 6).

We then analysed the strength of the links between the actors in the network and compared it with the results of the reconfiguration network, where the junction of each actor node in the figure (a) below represents the ratio of the number of linked edges between them to the number of possible linked edges in the network, and regularised the results of the reconfiguration network of the same size. We found that the links between R1 and R3 and R4 nodes in the linked network are more sparse than in the random network, while the links with R5 and R6 nodes are stronger than in the random network.

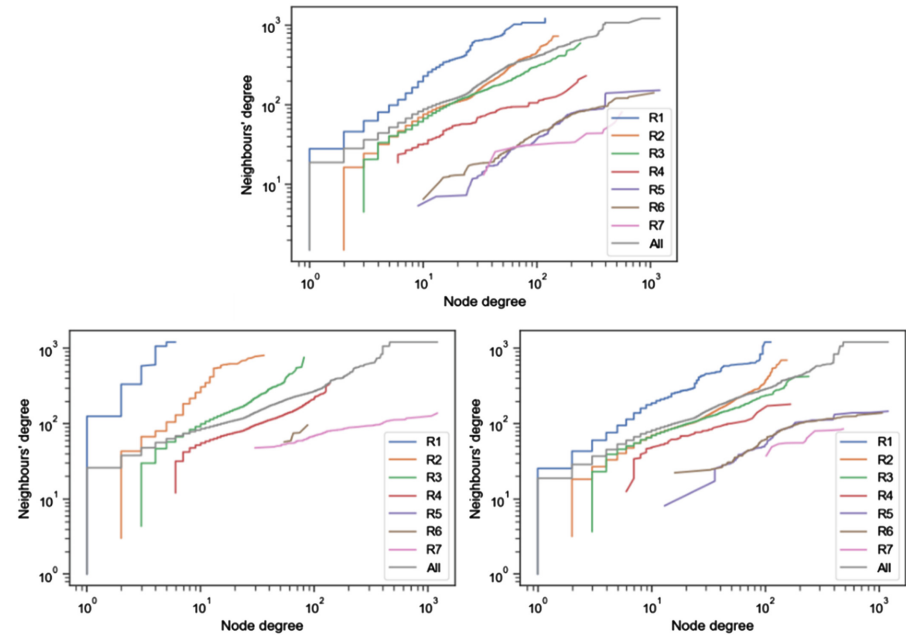


Fig. 6. Average degree of neighbor nodes of different role in 2002 wikilink network

That is, the general nodes in the community in the network are more connected to the central node of the community and less connected to the cross-community nodes, and the core nodes in the community of the explicit network are more connected to the bridge nodes between the communities, and it is through the nodes with high participation coefficients that the core nodes are connected to other nodes in the network. The number of R7 nodes is much less than in the random network, and thus the analysis of the strength of linked edges is not significant (Fig. 7).

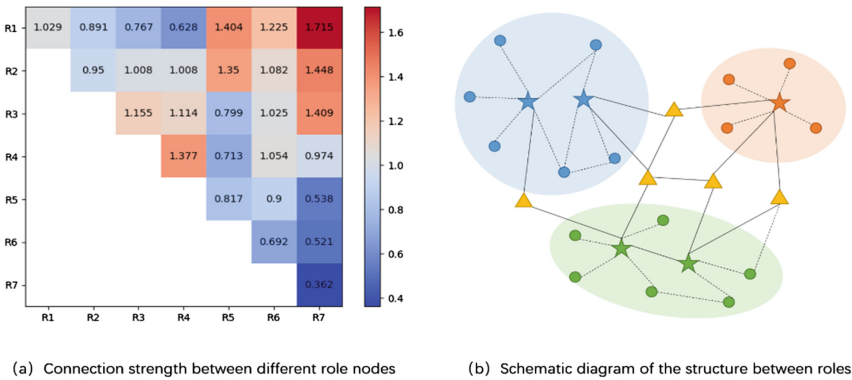


Fig. 7. Connection patterns between different role nodes in 2002 Wikilink network

Putting the above analysis together, we can derive a broad picture about the connectivity patterns between the different actors in the network, with a schematic representation of the connectivity patterns shown in (b) above. Where circles represent the general low-degree value nodes in the network, these are the most numerous nodes in the network and represent general concepts in the knowledge community. These nodes are more numerous than a random network of the same size and have little impact on the structure of the network but influence the size of the network. The richness of the knowledge content expands the body of knowledge in Wikipedia and is an important part of the knowledge network. The pentagram represents the core nodes of the network, which are the core concepts of the knowledge community, and form the knowledge skeleton of the community in which they are located by linking to the general entries that make up the content of the community. The triangles represent the very few cross-community nodes in the network, which are connected more strongly than other role nodes in the network and bridge the inter-community connections by linking to the core nodes within the community, playing an important role in the knowledge integration of the network.

7 Conclusion

This study constructed an annual Wikipedia network by data pre-processing and cleaning the raw data of the English Wikipedia link network from 2002–2018. The node roles and linking patterns of the network were analysed and it was found that the network is composed of a small number of highly core nodes and a large number of edge nodes populating it. Highly valued nodes are often located in different knowledge communities and are not tightly connected to each other. There is no obvious rich club phenomenon in the network, but rather connections are made through a plurality of club nodes across communities, which play an integral role in the transfer and circulation of information throughout the network.

The results of this study help us to gain insights into the characteristics and evolutionary patterns of Wikilink network, reveal the roles and information transfer mechanisms of key nodes in the network, and provide useful references for building more robust and efficient knowledge networks.

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